INDEX NUMBERS

INTRODUCTION

Historically, the first index was constructed in 1764 to compare the Italian price index in 1750 with the price level in 1500. Though originally developed for measuring the effect of change in prices, index numbers have today become one of the most widely used statistical devices and there is hardly any field where they are not used. Newspapers headline the fact that prices are going up or down, that industrial production is rising or falling, that imports are increasing or decreasing, that crimes are rising in a particular period compared to the previous period as disclosed by index numbers. They are used to feel the pulse of the economy and they have come to be used as indicators of inflationary or deflationary tendencies, In fact, they are described as 'barometers of economic activity', i.e., if one wants to get an idea as to what is happening to an economy, he should look to important indices like the index number of industrial production, agricultural production, business activity, etc.

Some prominent definitions of index numbers are given below:

1. 'Index numbers are devices for measuring differences in the magnitude of a group of related variables. —Croxton & Cowdert
2. "An index number is a statistical measure designed to show changes in a variable or a group of related variables with respect to time, geographic location or other characteristics such as income, profession, etc. —Spiegel
3. "In its simplest form an index number is the ratio of two index numbers expressed as a per cent. An index number is a statistical measure—a measure designed to show changes in one variable or in a group of related variables over time, or with respect to geographic location, or in terms of some other characteristics." —Patternson

Definition:

Index numbers are statistical devices designed to measure the relative change in the level of variable or group of variables with respect to time, geographical location etc.

In other words, these are the numbers which express the value of a variable at any given period called "***current period*** "as a percentage of the value of that variable at some standard period called "***base period***".

Here the variables may be

1. The price of a particular commodity like silver, iron or group of commodities like consumer goods, food, stuffs etc.
2. The volume of trade, exports, imports, agricultural and industrial production, sales in departmental store.
3. Cost of living of persons belonging to particular income group or profession etc.

Ex: suppose rice sells at Rs.9/kg at BBSR in 1995 as compare to Rs. 4.50/Kg in 1985, the index number of price in 1995 compared to 1985.

Therefore, the index number of price of rice in 1995 compared to 1985 is calculated as

$$\frac{Rs.9.00}{Rs.4.50} \times 100 = 200$$

This means there is a net increase of 100% in the price of rice in 1995as compared to 1985 **[the base year's index number is always treated as 100]**

Suppose, during the same period 1995 the rice sells at Rs. 12.00/kg in Delhi. There fore, the index number of price at Bhubaneswar compared to price at Delhi is $\frac{Rs.9.00}{Rs.12.00} \times 100 = 75$

This means there is a net decrease of 25% in the price of rice in 1995as compared to 1985

The above index numbers are called ***price index numbers***.

To take another example the production of rice in 1978 in Orissa was 44, 01,780 metric c tons compare to 36, 19,500 metric tons in 1971. So the index number of the quantity produced in 1978 compared to 1971 is

$$\frac{4401780}{3619500} \times 100 = 121.61$$

That means there is a net increase of 21.61% in production of rice in 1978 as compared to 1971.

The above index number is called ***'quantity index number'***

Univariate index: An index which is calculated from a single variable is called *univariate index.*

Composite index: An index which is calculated from group of variables is called *Composite index*

Characteristics of index numbers:

1. **Index numbers are specialized averages:**

As we know an average is a single figure representing a group of figures. How ever to obtain an average the items must be comparable. For example the average weight of man, woman and children of a certain locality has no meaning at all. Further more the unit of measurement must be same for all the items. However, this is not so with index numbers. Index numbers also one type of averages which shows in a single figure the change in two or more series of different items which can be expressed in different units. For example while constructing a consumer price index number the various items which are use in construction are divided into broad heads namely food, clothing, fuel, lighting, house rent, and miscellaneous which are expressed in different units.

2. **Index numbers measures the net change in a group of related variables:**

Since index numbers are essentially averages, they describe in one single figure the increase or decrease in a group of related variables under study. The group of variables may be prices of set of commodities, the volume of production in different sectors etc.

3. **Index numbers measure the effect of changes over period of time:**

Index numbers are most widely used for measuring changes over a period of time. For example we can compare the agricultural production, industrial production, imports, exports, wages etc in two different periods.

Uses of index numbers:

Index numbers are indispensable tools of economics and business analysis. Following are the main uses of index numbers.

1) **Index numbers are used as economic barometers:**

Index number is a special type of averages which helps to measure the economic fluctuations on price level, money market, economic cycle like inflation, deflation etc. G.Simpson and F.Kafka say that index numbers are today one of the most widely used statistical devices. They are used to take the pulse of economy and they are used as indicators of inflation or deflation tendencies. So index numbers are called economic barometers.

2) **Index numbers helps in formulating suitable economic policies and planning etc.**

Many of the economic and business policies are guided by index numbers. For example while deciding the increase of DA of the employees; the employer's have to depend primarily on the cost of living index. If salaries or wages are not increased according to the cost of living it leads to strikes, lock outs etc. The index numbers provide some guide lines that one can use in making decisions.

3) **They are used in studying trends and tendencies.**

Since index numbers are most widely used for measuring changes over a period of time, the time series so formed enable us to study the general trend of the phenomenon under study. For example for last 8 to 10 years we can say that imports are showing upward tendency.

4) **They are useful in forecasting future economic activity.**

Index numbers are used not only in studying the past and present workings of our economy but also important in forecasting future economic activity.

5) **Index numbers measure the purchasing power of money.**

The cost of living index numbers determine whether the real wages are rising or falling or remain constant. The real wages can be obtained by dividing the money wages by the corresponding price index and multiplied by 100. Real wages helps us in determining the purchasing power of money.

6) **Index numbers are used in deflating.**

Index numbers are highly useful in deflating i.e. they are used to adjust the wages for cost-of-living changes and thus transform nominal wages into real wages, nominal income to real income, nominal sales to real sales etc. through appropriate index numbers.

Classification of index numbers:

According to purpose for which index numbers are used are classified as below.

i) Price index number
ii) Quality index number
iii) Value index number
iv) Special purpose index number

Only price and quantity index numbers are discussed in detail. The others will be mentioned but without detail.

Price index number:

Price index number measures the changes in the price level of one commodity or group of commodities between two points of time or two areas.

Ex: Wholesale price index numbers
Retail price index numbers
Consumer price index numbers.

Quantity index number:

Quantity index numbers measure the changes in the volume of production, sales, etc in different sectors of economy with respect to time period or space.

Note: Price and Quantity index numbers are called *market index numbers.*

Problems in constructing index numbers:

Before constructing index numbers careful thought must be given into following problems

i. Purpose of index numbers.

An index number which is properly designed for a purpose can be most useful and powerful tool. Thus the first and the foremost problem are to determine the purpose of index numbers. If we know the purpose of the index numbers we can settle some related problems. For example, if the purpose of index number is to measure the changes in the production of steel, the problem of selection of items is automatically settled.

ii. Selection of commodities

After defining the purpose of index numbers, select only those commodities which are related to that index. For example if the purpose of an index is to measure the cost of living of low income group we should select only those commodities or items which are consumed by persons belonging to this group and due care should be taken not to include the goods which are utilized by the middle income group or high income group i.e. the goods like cosmetics, other luxury goods like scooters, cars, refrigerators, television sets etc.

iii. Selection of base period

The period with which the comparisons of relative changes in the level of phenomenon are made is termed as ***base period.*** The index for this period is always taken as 100. The following are the basic criteria for the choice of the base period.

i) The base period must be a normal period i.e. a period frees from all sorts of abnormalities or random fluctuations such as labor strikes, wars, floods, earthquakes etc.

ii) The base period should not be too distant from the given period. Since index numbers are essential tools in business planning and economic policies the base period should not be too far from the current period. For example for deciding increase in dearness allowance at present there is no advantage in taking 1950 or 1960 as the base, the comparison should be with the preceding year after which the DA has not been increased.

iii) Fixed base or chain base .While selecting the base a decision has to be made as to whether the base shall remain fixing or not i.e. whether we have fixed base or chain base. In the fixed base method the year to which the other years are compared is constant. On the other hand, in chain base method the prices of a year are linked with those of the preceding year. The chain base method gives a better picture than what is obtained by the fixed base method.

- **How a base is selected if a normal period is not available?**

  Ans: Some times it is difficult to distinguish a year which can be taken as a normal year and hence the average of a few years may be regarded as the value corresponding to the base year.

iv. Data for index numbers

The data, usually the set of prices and of quantities consumed of the selected commodities for different periods, places etc. constitute the raw material for the construction of index numbers. The data should be collected from reliable sources such as standard trade journals, official publications etc. for example for the construction of retail price index numbers, the price quotations for the commodities should be obtained from super bazaars, departmental stores etc. and not from wholesale dealers.

v.Selection of appropriate weights

A decision as to the choice of weights is an important aspect of the construction of index numbers. The problem arises because all items included in the construction are not of equal importance. So proper weights should be attached to them to take into account their relative importance. Thus there are two type of indices.

i) Un weighted indices- in which no specific weights are attached

ii) Weighted indices- in which appropriate weights are assigned to various items.

vi. Choice of average.

Since index numbers are specialized averages, a choice of average to be used in their construction is of great importance. Usually, the following averages are used.

i) A.M
ii) G.M
iii) Median

Among these averages ***G.M*** is the most appropriate average to be used. But in practice G.M is not used as often as A.M because of its computational difficulties.

vii. Choice of formula.

A large variety of formulae are available to construct an index number. The problem very often is that of selecting the appropriate formula. The choice of the formula would depend not only on the purpose of the index but also on the data available.

Methods of constructing index numbers:

A large number of formulae have been derived for constructing index numbers. They can be

1) Unweighted indices
   a) Simple aggregative method
   b) Simple average of relatives.
2) Weighted indices
   a) Weighted aggregative method
      i) Lasperey's method
      ii) Paasche's method
      iii) Fisher's ideal method
      iv) Dorbey's and Bowley's method
      v) Marshal-Edgeworth method
      vi) Kelly's method
   b) Weighted average of relatives

Unweighted indices:

i) Simple aggregative method:

This is the simplest method of constructing index numbers. When this method is used to construct a price index number the total of current year prices for the various commodities in question is divided by the total of the base year prices and the quotient is multiplied by 100.

`Symbolically $P_{01} = \frac{\sum P_1}{\sum P_0} \times 100$

Where: P_0 are the base year prices

P_1 are the current year prices

P_{01} is the price index number for the current year with reference to the base year.

Problem:

Calculate the index number for 1995 taking 1991 as the base for the following data

Commodity	Unit	Prices 1991 (P_0)	Prices 1995 (P_1)
A	Kilogram	2.50	4.00
B	Dozen	5.40	7.20
C	Meter	6.00	7.00
D	Quintal	150.00	200.00
E	Liter	2.50	3.00
Total		166.40	221.20

Price index number = $P_{01} = \frac{\sum P_1}{\sum P_0} \times 100 = \frac{221.20}{166.40} \times 100 = 132.93$

$\therefore$ There is a net increase of 32.93% in 1995 as compared to 1991.

Limitations:

There are two main limitations of this method

1. The units used in the prices or quantity quotations have a great influence on the value of index.
2. No considerations are given to the relative importance of the commodities.

ii) Simple average of relatives

When this method is used to construct a price index number, first of all price relatives are obtained for the various items included in the index and then the average of these relatives is obtained using any one of the averages i.e. mean or median etc.

When A.M is used for averaging the relatives the formula for computing the index is

$$P_{01} = \frac{1}{n} \sum \left(\frac{P_1}{P_0} \times 100 \right)$$

When G.M is used for averaging the relatives the formula for computing the index is

$$P_{01} = Anti\log \left[\frac{1}{n} \sum \log \left(\frac{P_1}{P_0} \times 100 \right) \right]$$

Where n is the number of commodities

$$\boxed{\text{price relative} = \frac{P_1}{P_0} \times 100}$$

Problem:

Calculate the index number for 1995 taking 1991 as the base for the following data

Commodity	Unit	Prices 1991 (P_0)	Prices 1995 (P_1)	$\frac{P_1}{P_0} \times 100$

A	Kilogram	50	70	$\frac{70}{50}\times 100$ = 140
B	Dozen	40	60	150
C	Meter	80	90	112.5
D	Quintal	110	120	109.5
E	Liter	20	20	100
Total				

Price index number = $P_{01} = \frac{1}{n}\sum\left(\frac{P_1}{P_0}\times 100\right) = \frac{1}{5}\sum 612 = 122.4$

$\therefore$ There is a net increase of 22.4% in 1995 as compared to 1991.

Merits:

1. It is not affected by the units in which prices are quoted
2. It gives equal importance to all the items and extreme items don't affect the index number.
3. The index number calculated by this method satisfies the unit test.

Demerits:

1. Since it is an unweighted average the importance of all items are assumed to be the same.
2. The index constructed by this method doesn't satisfy all the criteria of an ideal index number.
3. In this method one can face difficulties to choose the average to be used.

Weighted indices:

i) Weighted aggregative method:

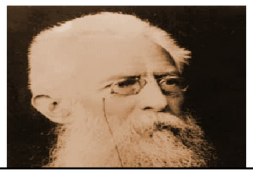
These indices are same as simple aggregative method. The only difference is in this method, weights are assigned to the various items included in the index.

There are various methods of assigning weights and consequently a large number of formulae for constructing weighted index number have been designed.

Some important methods are

i. Lasperey's method: This method is devised by Lasperey in year 1871.It is the most important of all the types of index numbers. In this method the base year quantities are taken weights. The formula for constructing Lasperey's price index number is

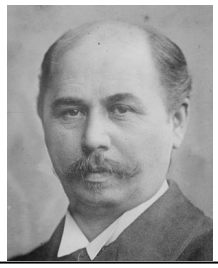
$$P_{01}^{La} = \frac{\sum p_1 q_0}{\sum p_0 q_0}\times 100$$



Ernst Louis Étienne Laspeyres (1834-1913) , Germany

ii. Paasche's method: In this method the current year quantities are taken as weights and the formula is given by

$$P_{01}^{Pa} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$$



Hermann Paasche (1851-1925)
Germany

iii. Fisher's ideal method: Fishers price index number is given by the G.M of the Lasperey's and Paasche's index numbers.

Symbolically

$$P_{01}^{F} = \sqrt{P_{01}^{La} P_{01}^{Pa}}$$

$$= \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100}$$

$$= \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

Sir Ronald Aylmer Fisher
(1890-1962) ,England

iv. Dorbey's and Bowley's method

Dorbey's and Bowley's price index number is given by the A.M of the Lasperey's and Paasche's index numbers.

Symbolically

$$P_{01}^{DB} = \frac{P_{01}^{La} + P_{01}^{Pa}}{2}$$

Quantity index numbers:

i. Laspeyre's quantity index number: Base year prices are taken as weights

$$Q_{01}^{La} = \frac{\sum q_1 p_0}{\sum q_0 p_0} \times 100$$

ii. Paasche's quantity index number: Current year prices are taken as weights

$$Q_{01}^{Pa} = \frac{\sum q_1 p_1}{\sum q_0 p_1} \times 100$$

iii. Fisher's ideal method: $Q_{01}^{F} = \sqrt{Q_{01}^{La} Q_{01}^{Pa}} = \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100$

Fisher's index number is called ideal index number. Why?

The Fisher's index number is called ideal index number due to the following characteristics.

1) It is based on the G.M which is theoretically considered as the best average of constructing index numbers.
2) It takes into account both current and base year prices as quantities.
3) It satisfies both time reversal and factor reversal test which are suggested by Fisher.
4) The upward bias of Lasperey's index number and downward bias of Paasche's index number are balanced to a great extent.

Example: Compute price index numbers for the following data by

(*i*) *Laspeyre's method,*
(*ii*) *Paasche's method,*
(*iii*) *Fisher's ideal method,*
(*iv*) *Dorbish-Bowley's method,*
(*v*) *Marshall-Edgeworth's method.*

Year	*Commodity A*		*Commodity B*		*Commodity C*	
	Price	*Quantity*	*Price*	*Quantity*	*Price*	*Quantity*
1980	*4*	*50*	*3*	*10*	*2*	*5*
1985	*10*	*45*	*6*	*8*	*3*	*4*

Base year : 1980
Price and quantity given in arbitrary units.

Calculation of Indices

Commodities	*1980*		*1985*					
	Price p_0	*Quantity* q_0	*Price* p_1	*Quantity* q_1	p_1q_0	p_0q_0	p_1q_1	p_0q_1
A	4	50	10	45	500	200	450	180
B	3	10	6	8	60	30	48	24
C	2	5	3	4	15	10	12	8
Total	—	—	—	—	575	240	510	212

(*i*) Laspeyre's method :

$$L_{01} = \frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times 100 = \frac{575}{240} \times 100 = 239.58$$

(*ii*) Paasche's method :

$$P_{01} = \frac{\Sigma p_1 q_1}{\Sigma p_0 q_1} \times 100 = \frac{510}{212} \times 100 = 240.57$$

(*iii*) Fisher's ideal method :

$$F_{01} = \sqrt{L_{01} \times P_{01}} = \sqrt{239.58 \times 240.57} = 240.07.$$

(*iv*) Dorbish-Bowley's method :

$$DB_{01} = \frac{L_{01} + P_{01}}{2}$$

$$= \frac{239.58 + 240.57}{2} \doteq 239.82.$$

Comparison of Lasperey's and Paasche's index numbers: -

In Lasperey's index number base year quantities are taken as the weights and in Paasche's index the current year quantities are taken as weights.

From the practical point of view Lasperey's index is often proffered to Paasche's for the simple reason that Laspeyre's index weights are the base year quantities and do not change from the year to the next. On the other hand, Paasche's index weights are the current year quantities, and in most cases these weights are difficult to obtain and expensive.

Lasperey's index number is said to be have upward bias because it tends to over estimate the price rise, where as the Paasche's index number is said to have downward bias, because it tends to under estimate the price rise.

When the prices increase, there is usually a reduction in the consumption of those items whose prices have increased. Hence using base year weights in the Lasperey's index, we will be giving too much weight to the prices that have increased the most and the numerator will be too large. Due to similar considerations, Paasche's index number using given year weights under estimates the rise in price and hence has down ward bias.

If changes in prices and quantities between the reference period and the base period are moderate, both Lasperey's and Paasche's indices give nearly the same values.

Demerit of Paasche's index number:

Paasche's index number, because of its dependence on given year's weight, has distinct disadvantage that the weights are required to be revised and computed for each period, adding extra cost towards the collection of data.

What are the desiderata of good index numbers?

Irving Fisher has considered two important properties which an index number should satisfy. These are tests of reversibility.

1. Time reversal test
2. Factor reversal test

If an index number satisfies these two tests it is said to be an ideal index number.

Main steps or problems in construction of cost-of-living index numbers

The following are the main steps in constructing a cost-of-living index number.

1. **Decision about the class of people for whom the index is meant**
   It is absolutely essential to decide clearly the class of people for whom the index is meant i.e. whether it relates to industrial workers, teachers, officers, labors, etc. Along with the class of people it is also necessary to decide the geographical area covered by the index, such as a city, or an industrial area or a particular locality in a city.
2. **Conducting family budget enquiry**
   Once the scope of the index is clearly defined the next step is to conduct a sample family budget enquiry i.e., we select a sample of families from the class of people for whom the index is

intended and scrutinize their budgets in detail. The enquiry should be conducted during a normal period i.e. a period free from economic booms or depressions. The purpose of the enquiry is to determine the amount; an average family spends on different items. The family budget enquiry gives information about the nature and quality of the commodities consumed by the people. The commodities are being classified under following heads

i) Food
ii) Clothing
iii) Fuel and Lighting
iv) House rent
v) miscellaneous

3. **Collecting retail prices of different commodities**
   The collection of retail prices is a very important and at the same time very difficult task, because such prices may vary from lace to place, shop to shop and person to person. Price quotations should be obtained from the local markets, where the class of people reside or from super bazaars or departmental stores from which they usually make their purchases.

Uses of cost-of-living index numbers:

1. Cost of living index numbers indicate whether the real wages are rising or falling. In other words they are used for calculating the real wages and to determine the change in the purchasing power of money.

$$\text{Purchasing power of money} = \frac{1}{\text{Cost of living index number}}$$

$$\text{R}eal\ Wages = \frac{\text{Money wages}}{\text{Cost of living index umbers}} \times 100$$

2. Cost of living indices are used for the regulation of D.A or the grant of bonus to the workers so as to enable them to meet the increased cost of living.
3. Cost of living index numbers are used widely in wage negotiations.
4. These index numbers also used for analyzing markets for particular kinds of goods.

Methods for construction of cost-of-living index numbers:

Cost of living index number can be constructed by the following formulae.

1) Aggregate expenditure method or weighted aggregative method
2) Family budget method or the method of weighted relatives

1) Aggregate expenditure method or weighted aggregative method

In this method the quantities of commodities consumed by the particular group in the base year are taken as weights. The formula is given by

$$\text{consumer price index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

Steps:

i) The prices of commodities for various groups for the current year is multiplied by the quantities of the base year and their aggregate expenditure of current year is obtained .i.e. $\sum p_1 q_0$

ii) Similarly obtain $\sum p_0 q_0$

iii) The aggregate expenditure of the current year is divided by the aggregate expenditure of the base year and the quotient is multiplied by 100.

Symbolically $\frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$

2) Family budget method or the method of weighted relatives

In this method cost of living index is obtained on taking the weighted average of price relatives, the weights are the values of quantities consumed in the base year i.e. $v = p_0 q_0$. Thus the consumer price index number is given by consumer price index = $\frac{\sum pv}{\sum v}$

Where $p = \frac{p_1}{p_o} \times 100$ for each item

$v = p_0 q_0$, value on the base year

Note: It should be noted that the answer obtained by applying the aggregate expenditure method and family budget method shall be same.

Example: Construct the consumer price index number for 2007 on the basis of 2006 from the following data using (i) the aggregate expenditure method, and (ii) the family budget method.

Commodity	*Quantity consumed in 2006*	*Units*	*Price in 2006*		*Price in 2007*	
			Rs.	*Paise*	*Rs.*	*Paise*
A	6 Quintals	Quintal	5	75	6	0
B	6 Quintals	Quintal	5	0	8	0
C	1 Quintals	Quintal	6	0	9	0
D	6 Quintals	Quintal	8	0	10	0
E	4 Kg.	Kg.	2	0	1	50
F	1 Quintals	Quintal	20	0	15	0

Solution. COMPUTATION OF CONSUMER PRICE INDEX NUMBER FOR 2007 (Base 2006 = 100) BY THE AGGREGATE EXPENDITURE METHOD

Commo-dity	*Quantities consumed*	*Unit*	*Price in 2006* p_0	*Price in 2007* p_1	$p_1 q_0$	$p_0 q_0$
A	6 Qtl.	Qtl.	5.75	6.00	36.00	34.50
B	6 Qtl.	"	5.00	8.00	48.00	30.00
C	1 Qtl.	"	6.00	9.00	9.00	6.00
D	6 Qtl.	"	8.00	10.00	60.00	48.00
E	4 Kg.	Kg.	2.00	1.50	6.00	8.00
F	1 Qtl.	Qtl.	20.00	15.00	15.00	20.00
					$\Sigma p_1 q_0 = 174$	$\Sigma p_0 q_0 = 146.5$

Consumer Price Index $= \frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times 100 \; \frac{174}{146.5} \times 100 = 118.77$

CONSTRUCTON OF CONSUMER PRICE INDEX NUMBER FOR 2007
(Base 2006 = 100) BY THE FAMILY BUDGET METHOD

Commo-dity	*Quantities consumed* q_0	*Unit*	*Price in 2006* p_0	*Price in 2007* p_1	$\frac{p_1}{p_0} \times 100$ P	p_0q_0 V	PV
A	6 Qtl:	Qtl.	5.75	6.0	104.34	34.5	3,600
B	6 Qtl.	Qtl.	5.00	8.0	160.00	30.0	4,800
C	1 Qtl.	Qtl.	6.00	9.0	150.00	6.0	900
D	6 Qtl.	Qtl.	8.00	10.0	125.00	48.0	6,000
E	4 Kg.	Kg.	2.00	1.5	75.00	8.0	600
F	1 Qtl.	Qtl.	20.00	15.0	75.00	20.0	1,500
						ΣV =146.5	ΣPV = 17,400

$$\text{Consumer Price Index} = \frac{\Sigma PV}{\Sigma V} = \frac{17.400}{146.5} = 118.77$$

Thus, the answer is the same by both the methods. However, the reader should prefer the aggregate expenditure method because it is far easier to apply compared to the family budget method.

Possible errors in construction of cost-of-living index numbers:

Cost of living index numbers or its recently popular name consumer price index numbers are not accurate due to various reasons.

2. Errors may occur in the construction because of inaccurate specification of groups for whom the index is meant.
3. Faulty selection of representative commodities resulting out of unscientific family budget enquiries.
4. Inadequate and unrepresentative nature of price quotations and use of inaccurate weights
5. Frequent changes in demand and prices of the commodity
6. The average family might not be always a representative one.

Problems or steps in construction of wholesale price index numbers (WPI):

Index numbers are the best indicators of the economic progress of a community, a nation and the world as a whole. Wholesale price index numbers can also be constructed for different economic activities such as Indices of Agricultural production, Indices of Industrial production, Indices of Foreign Trade etc. Besides some International organizations like the United Nations Organization, the F.A.O. of the U.N., the World Bank and International Labour Organization, there are a number of organizations in the country who publish index numbers on different aspects. These are (a) Ministry of Food and Agriculture, (b) Reserve Bank of India, (c) Central Statistical Organization, (d) Department of Commercial Intelligence and Statistics, (e) Labour Bureau, (f) Eastern Economist. The Central Statistical Organization of the Government of India publishes a Monthly Abstract of' Statistics which contains All India index

numbers of Wholesale Prices (Revised series: Base year 1981-82) both commodity-wise and also for the aggregate.

i. Purpose or object of index numbers.

A wholesale price index number which is properly designed for a purpose can be most useful and powerful tool. Thus, the first and the foremost problem are to determine the purpose of index numbers. If we know the purpose of the index numbers we can settle some related problems.

ii. Selection of commodities

Representative items should be taken into consideration. The items may be grouped into relatively homogeneous heads to make the calculation. The construction of WPI of a region or country we may group the commodities as (1) Primary Articles — (a) Food Articles (b) Non-food Articles (c) Minerals (ii) Fuel. Power, Light and Lubricants (iii) Manufactured Products (iv) Chemicals and Chemical Products (v) Machinery and Machine Equipments (vi) Other Miscellaneous Manufacturing Industries.

iii. Selection of base period

1. The base period must be a normal period i.e. a period frees from all sorts of abnormalities or random fluctuations such as labor strikes, wars, floods, earthquakes etc.
2. The base period should not be too distant from the given period. Since index numbers are essential tools in business planning and economic policies the base period should not be too far from the current period. For example, for deciding increase in dearness allowance at present there is no advantage in taking 1950 or 1960 as the base, the comparison should be with the preceding year after which the DA has not been increased.
3. Fixed base or chain base. While selecting the base a decision has to be made as to whether the base shall remain fixing or not i.e. whether we have fixed base or chain base. In the fixed base method, the year to which the other years are compared is constant. On the other hand, in chain base method the prices of a year are linked with those of the preceding year. The chain base method gives a better picture than what is obtained by the fixed base method.

iv. Data for index numbers

The data, usually the set of prices and of quantities consumed of the selected commodities for different periods, places etc. constitute the raw material for the construction of wholesale rice index numbers. The data should be collected from reliable sources such as standard trade journals, official publications etc.

v. Selection of appropriate weights

A decision as to the choice of weights is an important aspect of the construction of index numbers. The problem arises because all items included in the construction are not of equal importance. So proper weights should be attached to them to take into account their relative importance. Thus there are two type of indices.

1. Un weighted indices- in which no specific weights are attached
2. Weighted indices- in which appropriate weights are assigned to various items.

vi. Choice of average.

Since index numbers are specialized averages, a choice of average to be used in their construction is of great importance. Usually, the following averages are used.

iv) A.M

v) G.M

vi) Median

Among these averages ***G.M*** is the appropriate average to be used. But in practice G.M is not used as often as A.M because of its computational difficulties.

vii. Choice of formula.

The selection of a formula along with a method of averaging depends on data at hand and purpose for which it is used. Different formulae developed for the purpose have already been discussed in earlier sections.

Wholesale price index numbers (Vs) consumer price index numbers:

1. The wholesale price index number measures the change in price level in a country as a whole. For example, economic advisors index numbers of wholesale prices. Where as cost-of-living index numbers measures the change in the cost of living of a particular class of people stationed at a particular place. In this index number we take retail price of the commodities.
2. The wholesale price index number and the consumer price index numbers are generally different because there is lag between the movement of wholesale prices and the retail prices.
3. The retail prices required for the construction of consumer price index number increased much faster than the wholesale prices i.e. there might be erratic changes in the consumer price index number unlike the wholesale price index numbers.
4. The method of constructing index numbers in general the same for wholesale prices and cost of living. The wholesale price index number is based on different weighting systems and the selection of commodities is also different as compared to cost-of-living index number

Importance and methods of assigning weights:

The problem of selecting suitable weights is quite important and at the same time quite difficult to decide. The term weight refers to the relative importance of the different items in the construction of the index. Generally various items say wheat, rice, kerosene, clothing etc. included in the index are not of equal importance, proper weights should be attached to them to take into their relative importance. Thus, there are two types of indices.

1) Unweighted indices – in which no specific weights are attached to various commodities.
2) Weighted indices – in which appropriate weights are assigned to various commodities.

The Unweighted indices can be interpreted as weighted indices by assuming the corresponding weight for each commodity being unity. But actually, the commodities included in the index are all not of equal importance. Therefore, it is necessary to adopt some suitable method of weighting, so that arbitrary and haphazard weights may not affect the results.

There are two methods of assigning weights.

i) Implicit weighting

ii) Explicit weighting

In *implicit weighting,* a commodity or its variety is included in the index a number of times. For example, if wheat is to be given in an index twice as much times as rice then the weight of wheat is two. Where as in *explicit weighting* two types of weights can be assigned. i.e. quantity weights or value weights.

A quantity weight symbolized by q means the amount of commodity produced, distributed or consumed in some time period. A value weight in the other hand combines price with quantity produced, distributed or consumed and is denoted by v=pq.

For example, quantity weights are used in the method of weighted aggregative like Lasperey's, Paasche's index numbers and value weights are used in the method of weighted average of price relatives.

Limitations or demerits of index numbers:

Although index numbers are indispensable tools in economics, business, management etc, they have their limitations and proper care should be taken while interpreting them. Some of the limitations of index numbers are

1. Since index numbers are generally based on a sample, it is not possible to take into account each and every item in the construction of index.
2. At each stage of the construction of index numbers, starting from selection of commodities to the choice of formulae there is a chance of the error being introduced.
3. Index numbers are also special type of averages, since the various averages like mean, median, G.M have their relative limitations, their use may also introduce some error.
4. None of the formulae for the construction of index numbers is exact and contains the so-called *formula error*. For example, Lasperey's index number has an upward bias while Paasche's index has a downward bias.
5. An index number is used to measure the change for a particular purpose only. Its misuse for other purpose would lead to unreliable conclusions.
6. In the construction of price or quantity index numbers it may not be possible to retain the uniform quality of commodities during the period of investigation.